

CS4084

Quantum Computing

Week 02 – Lecture 03

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Quick Review: What We Covered Last Week

- Introduction to **Quantum Computing**
- Difference between **Classical Physics** and **Quantum Physics**
- Concept of **Qubits** and **Superposition**
- **Bloch Sphere** representation of a single qubit

Key Message: Quantum information behaves fundamentally differently from classical information.

Quantum Information and Computation: Roadmap

- We begin by understanding how **quantum information works for a single system** in isolation (a single qubit).
- After that, we **extend these ideas to multiple quantum systems**, leading to:
 - Multi-qubit states
 - Entanglement and correlations
- Studying single systems first helps build intuition before moving to more complex quantum behavior.

From Classical to Quantum Information

- Before exploring quantum information, we must first understand:
 - What **classical information** is
 - How it is **stored, processed, and measured**
- **Quantum computing is an extension of classical computing**, not a replacement.
- At the **mathematical level**, classical and quantum computing share similar concepts:
 - States
 - Operations
 - Measurements
- Key differences arise due to quantum effects such as:
 - Superposition
 - Probabilistic measurement
 - Entanglement

Description of Quantum Information

Simplified description (this unit)

- Simpler and typically learned first
- Quantum states represented by **vectors**; operations are represented by **unitary matrices**
- Sufficient for an understanding of most quantum algorithms

General description (covered in a later unit)

- More general and more broadly applicable
- Quantum states represented by **density matrices**; allows for a more general class of measurements and operations

Classical Information Processing

Consider a physical system that stores information: let us call it X .

Assume X can be in one of a finite number of **classical states** at each moment. Denote this classical state set by Σ .

Examples

- If X is a bit, then its classical state set is $\Sigma = \{0, 1\}$.
- If X is a six-sided die, then $\Sigma = \{1, 2, 3, 4, 5, 6\}$.
- If X is a switch on a standard electric fan, then perhaps $\Sigma = \{\text{high, medium, low, off}\}$.

Mathematical Description of Classical Information

For example, if X is a bit, then perhaps it is in the classical state 0 with probability $3/4$ and in the classical state 1 with probability $1/4$. This is a **probabilistic state** of X .

$$\Pr(X = 0) = \frac{3}{4} \quad \text{and} \quad \Pr(X = 1) = \frac{1}{4}$$

A succinct way to represent this probabilistic state is by a **column vector**:

$$\begin{pmatrix} \frac{3}{4} \\ \frac{1}{4} \end{pmatrix} \begin{array}{l} \leftarrow \text{entry corresponding to 0} \\ \leftarrow \text{entry corresponding to 1} \end{array}$$

This vector is a **probability vector**:

- All entries are nonnegative real numbers.
- The sum of the entries is 1.

Dirac Notation (1st Part)

Let Σ be any classical state set, and assume the elements of Σ have been placed in correspondence with the integers $1, \dots, |\Sigma|$.

We denote by $|\alpha\rangle$ the **column vector** having a 1 in the entry corresponding to $\alpha \in \Sigma$, with 0 for all other entries.

Example 1

If $\Sigma = \{0, 1\}$, then

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Standard deck of Playing Cards Mathematical Examples

Let Σ be any classical state set, and assume the elements of Σ have been placed in correspondence with the integers $1, \dots, |\Sigma|$.

We denote by $|\alpha\rangle$ the **column vector** having a 1 in the entry corresponding to $\alpha \in \Sigma$, with 0 for all other entries.

Example 2

If $\Sigma = \{\clubsuit, \diamondsuit, \heartsuit, \spadesuit\}$, then we might choose to order these states like this:
 $\clubsuit, \diamondsuit, \heartsuit, \spadesuit$. This yields

$$|\clubsuit\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad |\diamondsuit\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad |\heartsuit\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad |\spadesuit\rangle = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$$

Alternative Way to Represent Vectors - Standard Basis vectors

Vectors of this form are called *standard basis vectors*. Every vector can be expressed uniquely as a linear combination of standard basis vectors.

Example

$$\begin{pmatrix} \frac{3}{4} \\ \frac{1}{4} \end{pmatrix} = \frac{3}{4} |0\rangle + \frac{1}{4} |1\rangle$$

Measuring probabilistic states in Classical Domain

What happens if we **measure** a system X while it is in some probabilistic state?

We see a **classical state**, chosen at random according to the probabilities.

Suppose we see the classical state $\alpha \in \Sigma$.

This changes the probabilistic state of X (from our viewpoint): having recognized that X is in the classical state α , we now have

$$\Pr(X = \alpha) = 1$$

Deterministic operations

Every function $f : \Sigma \rightarrow \Sigma$ describes a **deterministic operation** that transforms the classical state α into $f(\alpha)$, for each $\alpha \in \Sigma$.

Given any function $f : \Sigma \rightarrow \Sigma$, there is a (unique) matrix M satisfying

$$M |\alpha\rangle = |f(\alpha)\rangle \quad (\text{for every } \alpha \in \Sigma)$$

This matrix has exactly one 1 in each column, and 0 for all other entries:

$$M(b, \alpha) = \begin{cases} 1 & b = f(\alpha) \\ 0 & b \neq f(\alpha) \end{cases}$$

The action of this operation is described by **matrix-vector multiplication**:

$$v \mapsto Mv$$

Mathematical Examples of Deterministic operations

Example

For $\Sigma = \{0, 1\}$, there are four functions of the form $f : \Sigma \rightarrow \Sigma$:

a	$f_1(a)$	a	$f_2(a)$	a	$f_3(a)$	a	$f_4(a)$
0	0	0	0	0	1	0	1
1	0	1	1	1	0	1	1

Here are the matrices corresponding to these functions:

$$M_1 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \quad M_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad M_3 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad M_4 = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$$

Visual Comparison: Classical vs Quantum Information

Classical Information

- Bit: 0 or 1
- Definite state
- Deterministic operations

Bit		
0	OR	1

Quantum Information

- Qubit: $\alpha|0\rangle + \beta|1\rangle$
- Superposition of states
- Probabilistic measurement

Qubit
$ 0\rangle + 1\rangle$

Summary: Key Takeaways

- **Classical information** is based on definite states (0 or 1) and deterministic operations.
- Classical systems use probability only due to **lack of knowledge**, not because the system itself is uncertain.
- **Quantum information** is fundamentally different:
 - States can exist in **superposition**
 - Measurement outcomes are **inherently probabilistic**
 - Information is represented using **complex amplitudes**
- Understanding classical information first helps us clearly see what is **new and powerful** in quantum computing.

Next: We will move from classical states to **quantum states (qubits)** and their mathematical description.